

//Static Functions class file

```
public class Static_Function
{
    private static int x,y;
    public static void add(int x,int y)
    {
        int add;
        add=x+y;
        System.out.println("Addition is = "+add);
    }
    public static void sub(int x,int y)
    {
        int sub;
        sub=x-y;
        System.out.println("Subtraction is = "+sub);
    }
    public static void mul(int x,int y)
    {
        int mul;
        mul=x*y;
        System.out.println("Multiplication is = "+mul);
    }
    public static void div(int x,int y)
    {
        int div;
        div=x/y;
        System.out.println("Division is = "+div);
    }
}
```

//Static Function Test File

```
import java.util.Scanner;
public class Static_Test
{
    public static void main(String[] args)
    {
        int x,y;
        Scanner input=new Scanner(System.in);
        Static_Function sf=new Static_Function();
        System.out.println("Enter the values of X and Y = ");
        x=input.nextInt();
        y=input.nextInt();

        sf.add(x,y);
        sf.sub(x,y);
        sf.mul(x,y);
        sf.div(x,y);
    }
}
```

OUTPUT

```
C:\Users\lenovo\java\Practicals\6,7,8>javac Static_Function.java
C:\Users\lenovo\java\Practicals\6,7,8>javac Static_Test.java
C:\Users\lenovo\java\Practicals\6,7,8>java Static_Test
Enter the values of X and Y =
8 4
Addition is = 12
Subtraction is = 4
Multiplication is = 32
Division is = 2
C:\Users\lenovo\java\Practicals\6,7,8>_
```

//Number of Objects count

```
public class countobj {
    private static int count ;
    public countobj()
    {
        count++;
    }
    public static void main(String args[])
    {
        countobj ob1 = new countobj();
        countobj ob2 = new countobj();
        countobj ob3 = new countobj();
        countobj ob4 = new countobj();
        countobj ob5 = new countobj();
        System.out.println("Total Number of Objects = "
            +countobj.count);
    }
}
```

OUTPUT

```
C:\Users\lenovo\java\Practicals\6,7,8>javac countobj.java
C:\Users\lenovo\java\Practicals\6,7,8>java countobj
Total Number of Objects = 5
C:\Users\lenovo\java\Practicals\6,7,8>
```

// Room class using constructors

```
public class RoomUC
{
    protected int length,width;
    RoomUC(int a,int b)
    {
        length=a;
        width=b;
    }
    public void setData(int l, int w)
    {
        length=l;
        width=w;
    }
    public int area()
    {
        return(length*width);
    }
}
class Kitchen extends RoomUC
{
    private int height;
    Kitchen(int a,int b,int c)
    {
        super(a,b);
        height=c;
    }
    public void setHeight(int h)
    {
        height=h;
    }
    int volume()
    {
        return(length*width*height);
    }
}
```

//Room Test class using constructors

```
public class RoomTest_UC
{
    public static void main(String[] args)
    {
        Kitchen k=new Kitchen(20,30,40);

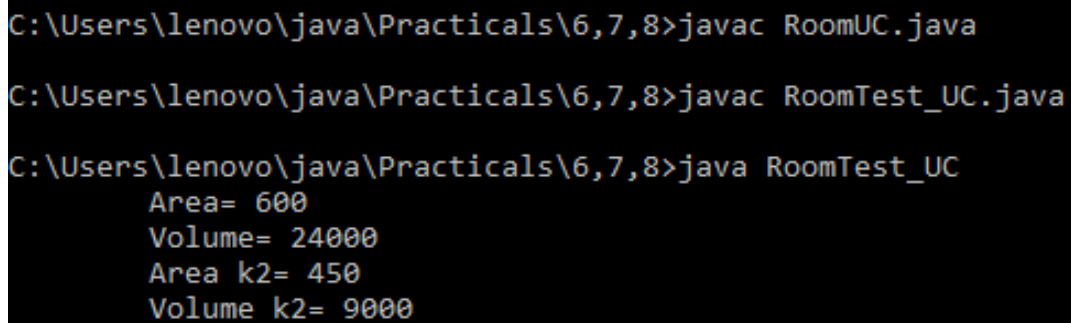
        int area=k.area();
        int vol=k.volume();
        System.out.println("\tArea= "+area);
        System.out.println("\tVolume= "+vol);

        Kitchen k2=new Kitchen(30,15,20);

        int area2=k2.area();
        int vol2=k2.volume();
        System.out.println("\tArea k2= "+area2);
        System.out.println("\tVolume k2= "+vol2);

    }
}
```

OUTPUT



```
C:\Users\lenovo\java\Practicals\6,7,8>javac RoomUC.java
C:\Users\lenovo\java\Practicals\6,7,8>javac RoomTest_UC.java
C:\Users\lenovo\java\Practicals\6,7,8>java RoomTest_UC
    Area= 600
    Volume= 24000
    Area k2= 450
    Volume k2= 9000
```

//Function Overriding class file

```
class A
{
    int i,j;
    A(int a,int b)
    {
        i=a;
        j=b;
    }
    public void show()
    {
        System.out.println("i = "+i+"\nj = "+j);
    }
}
class B extends A
{
    int k;
    B(int a,int b,int c)
    {
        super(a,b);
        k=c;
    }
    public void show()
    {
        super.show();
        System.out.println("k = "+k);
    }
}
```

//Function overriding test file

```
public class ABTest
{
    public static void main(String[] args)
    {
        int a,b,c;
        B b1=new B(10,20,30);
        b1.show();
    }
}
```

OUTPUT

```
C:\Users\lenovo\java\Practicals\6,7,8>javac A.java
C:\Users\lenovo\java\Practicals\6,7,8>javac ABTest.java
C:\Users\lenovo\java\Practicals\6,7,8>java ABTest
i = 10
j = 20
k = 30
```


// Figure class file

```
abstract class Figure
{
    protected double dim1;
    protected double dim2;

    Figure(double a,double b)
    {
        dim1=a;
        dim2=b;
    }
    abstract double area();
}
class Rectangle extends Figure
{
    Rectangle(double a,double b)
    {
        super(a,b);
    }
    double area()
    {
        System.out.println("Inside area of Rectangle");
        return dim1*dim2;
    }
}

class Triangle extends Figure
{
    Triangle(double a,double b)
    {
        super(a,b);
    }
    double area()
    {
        System.out.println("Inside area of Triangle");
        return dim1*dim2/2;
    }
}

class Circle extends Figure
{
    Circle(double a,double b)
    {
        super(a,b);
    }
    double area()
    {
        System.out.println("Inside area of circle");
        return 3.14*dim1*dim1;
    }
}
```


//Abstract Test File

```
public class Example_AbstractClassTest
{
    public static void main(String[] args)
    {
        Rectangle r=new Rectangle(10,20);
        Triangle t=new Triangle(10,8);
        Circle c=new Circle(10,0);
        Figure f;

        f=r;
        System.out.println("Area of rectangle="+f.area());
        System.out.println("");
        f=t;
        System.out.println("Area of triangle="+f.area());
        f=c;
        System.out.println("Area of circle="+f.area());
    }
}
```

OUTPUT

```
C:\Users\lenovo\java\Practicals\6,7,8>javac Figure.java
C:\Users\lenovo\java\Practicals\6,7,8>javac FigureTest_Abstract.java
C:\Users\lenovo\java\Practicals\6,7,8>java FigureTest_Abstract
    Inside area of Rectangle
    Area of rectangle=200.0
    Inside area of Triangle
    Area of triangle=40.0
    Inside area of circle
    Area of circle=314.0
```